
Spatial Patterns in Indian District Night-Light Growth

Descriptive statistics, spatial econometrics, and cluster analysis

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This report studies growth in satellite-observed nighttime light across 640 Indian Census 2011 districts. Results describe associations in remotely sensed light; they are not estimates of district GDP or causal spatial spillovers.

Abstract

This study analyzes 640 Indian Census 2011 districts using SHRUG v2.2 nighttime lights. The main comparison uses VIIRS observations from 2013 to 2021 and fixed 2011 population to normalize district size. Lower initially lit districts show faster subsequent light growth in a conditional model, while growth is strongly spatially clustered. Spatial lag, spatial error, and spatial Durbin models separate forms of dependence. Local Moran analysis and a graph-constrained Ward typology describe regional clusters. These are associations in satellite light, not district GDP or causal spillovers.

Key results

Headline estimates

Measure	Estimate
Mean annual 2013–2021 log-light growth	0.0682
Conditional OLS initial-light coefficient	−0.0457 (HC3 SE 0.00344)
Global Moran’s I for growth	0.654 (permutation $p = 0.001$)
Spatial Durbin ρ	0.3253
Spatial Durbin residual Moran’s I	−0.0037 ($p = 0.935$)
FDR-significant high–high / low–low districts	37 / 37

Interpretation. Light is normalized by each district’s 2011 population, held fixed in every year. It is therefore baseline-population-normalized light, not contemporaneous per-capita income.

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1 Research question and data

The first project in the supplied brief asks whether less initially lit Indian districts grew faster, whether growth forms spatial clusters, and whether nearby outcomes or characteristics are associated with district growth. The unit is the Census 2011 district, held fixed over time. Night lights are an imperfect economic-activity proxy: electrification, lighting technology, and settlement patterns also matter [2].

The source tables are SHRUG v2.2 district VIIRS annual sum, calibrated DMSP district light, the 2011 Population Census Abstract, a population/land-area key, and district polygons [3, 1]. The raw VIIRS table covers 2012–2023, but published metadata describes 2012–2021; the documented range is used here. VIIRS and DMSP are never spliced.

1.1 Data audit

Table 1: Data-integrity checks.

Check	Result
District census and area rows	640 each; population values agree
VIIRS derived panel	12,800 rows = $640 \times 10 \text{ years} \times 2 \text{ masks}$
DMSP derived panel	12,800 rows = $640 \times 20 \text{ years}$
Polygon match	640 exact keys; one unlabeled district-000 placeholder removed
Main endpoint sample	640 districts, no missing covariates, positive endpoint VIIRS light

1.2 Variable construction

For district i and year t , normalized light, log light, and annual growth are defined as

$$L_{it} = \frac{1,000 \times \text{VIIRS_sum}_{it}}{\text{population}_{i,2011}}, \quad (1)$$

$$z_{it} = \log(L_{it}), \quad (2)$$

$$g_i = \frac{z_{i,2021} - z_{i,2013}}{8}. \quad (3)$$

The fixed denominator means that the growth rate equals growth of total district light. Baseline controls are log 2011 density, adult literacy, agricultural main-worker share, Scheduled Caste/Tribe share, and state indicators.

The principal log comparison starts in 2013 because the 2012 VIIRS observations include zeros. For DMSP, concurrent calibrated satellite measurements are averaged within district-year. The historical log check begins in 2000, the first year positive for every district.

2 Numerical and visual description

Table 2: District-level descriptive statistics.

District variable	Mean	Median	P10	P90
VIIRS per 1,000 baseline residents, 2013	7.689	6.068	1.641	15.259
VIIRS per 1,000 baseline residents, 2021	11.209	9.829	4.653	18.272
Annual log-light growth	0.0682	0.0588	−0.0004	0.1478
2011 density	994.0	409.8	137.7	1,251.8
2011 literacy share	0.723	0.722	0.590	0.854
2011 agricultural worker share	0.533	0.591	0.197	0.773

Mean normalized light rises 45.8% from 2013 to 2021; the median rises 62.0%. Mean annual log growth is 0.0682, and 10.5% of districts have negative endpoint growth. The distribution is right-skewed. The annual series has a noticeable 2016–2017 step, which could reflect illumination or processing as well as activity.

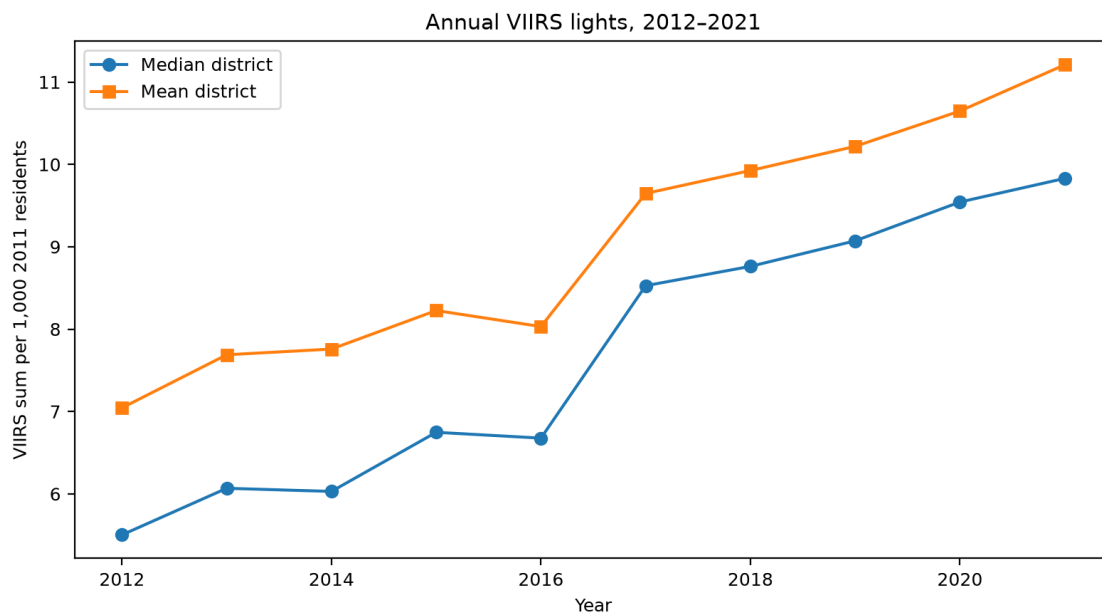


Figure 1: VIIRS district mean and median light per 1,000 fixed 2011 residents.

3 Distribution and spatial clustering

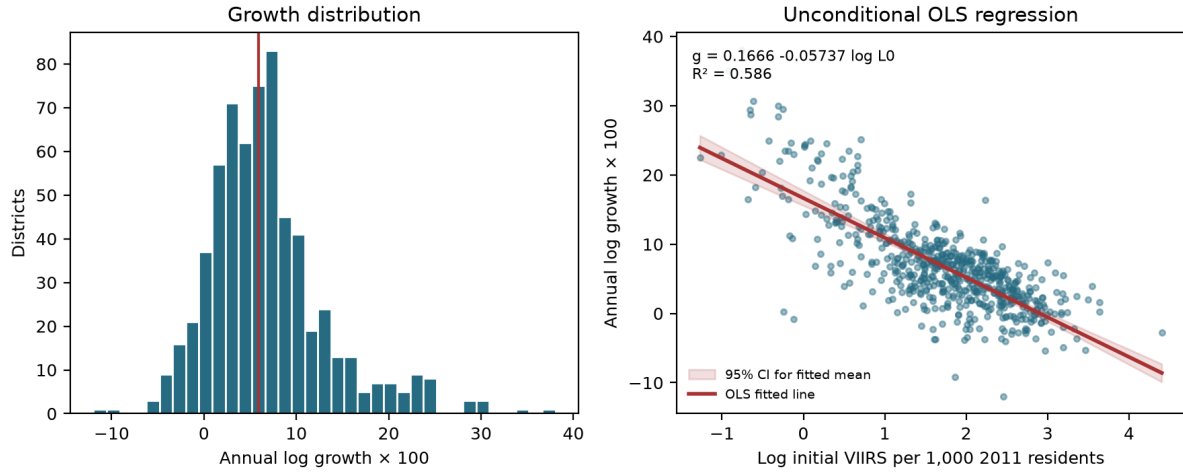


Figure 2: Growth distribution and unconditional OLS line with a 95% confidence band for the fitted mean.

The fitted unconditional line is

$$\hat{g} = 0.16662 - 0.05737 \log(\text{initial normalized VIIRS light}), \quad R^2 = 0.586.$$

The negative slope is an association in annual log-light growth, not a line for GDP growth. The conditional fitted relation instead has an initial-light slope of -0.04569 after controls and state effects (Section 6).

Global Moran's I compares district deviations with the weighted deviations of Queen-contiguous neighbors. The Queen graph has 4.14 neighbors per district on average, 34 disconnected components, and 12 districts with no contiguous neighbor. Those 12 retain zero rows in $\mathbf{W}$; no distant boundary is invented. An eight-nearest-neighbor graph is a sensitivity test.

Table 3: Global spatial-autocorrelation tests.

Moran test	I	Two-sided permutation p
Initial normalized light	0.6124	0.001
Annual growth	0.6544	0.001
Conditional OLS residual	0.1693	0.001
Spatial Durbin residual	-0.0037	0.935

Local Moran statistics use 999 permutations and Benjamini–Hochberg false discovery rate control at 5% over the 628 districts with a Queen neighbor. Results are 37 high–high, 37 low–low, two low–high, 552 not significant, and 12 with no contiguous neighbor. High–high examples include districts in Bihar and Manipur. Spatial clustering does not by itself establish diffusion.

4 Where growth clusters occur

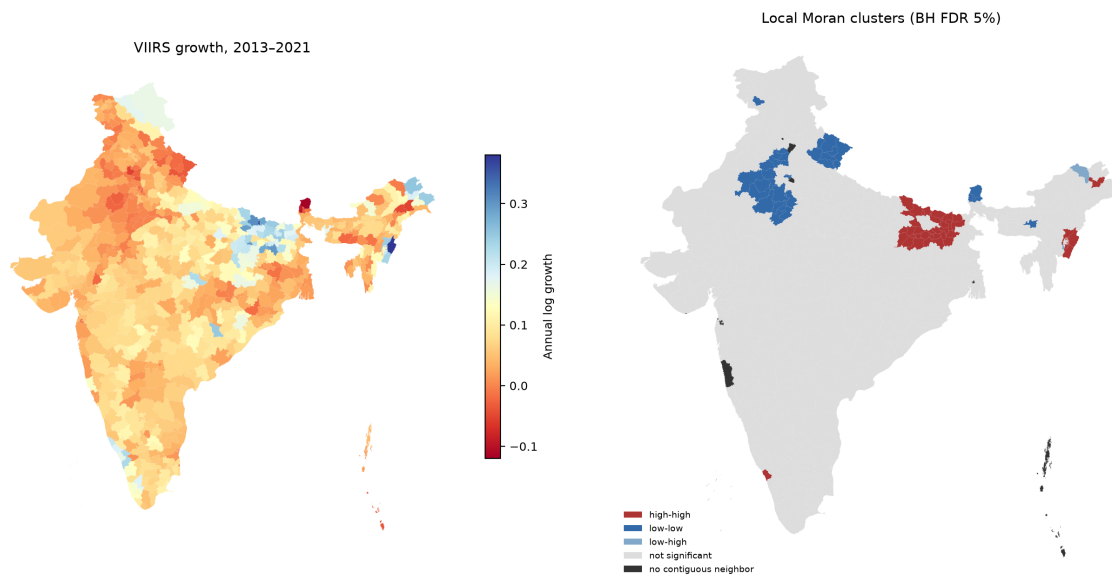


Figure 3: Annual 2013–2021 growth and Local Moran classes after FDR correction. Dark gray marks no-neighbor districts.

The map shows broad geographic heterogeneity, including fast-growing clusters in parts of Bihar and the northeast. Neighbor definitions matter where the source polygons have gaps or inaccuracies; the supplied SHRUG map documentation cautions about boundary precision.

4.1 Graph-constrained Ward grouping

We standardize five district features: initial log light, annual growth, log density, literacy, and agricultural-employment share. Ward agglomerative clustering may merge districts only along a symmetric eight-nearest-neighbor graph. We compare two through six groups with the silhouette score. Three groups score highest (0.158), and each is connected in the graph. The low score signals modest separation, so these are a descriptive typology, not sharply defined economic regions. Growth is an input to clustering and cannot be used as an independent validation.

5 Spatially constrained district typology

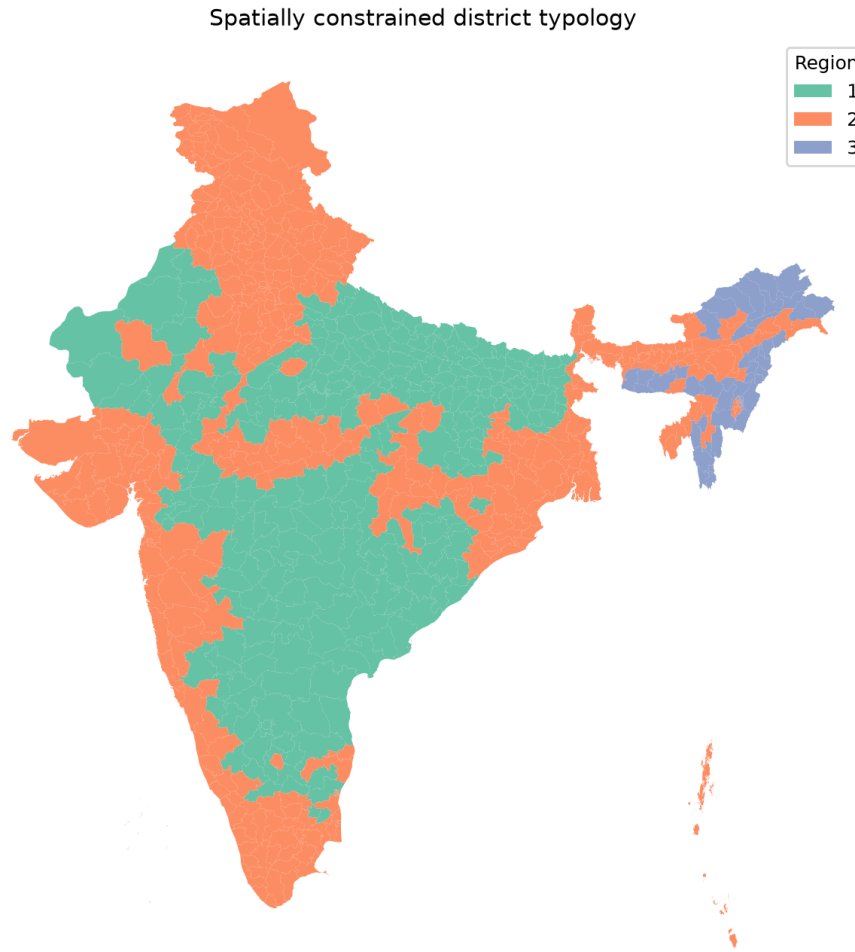


Figure 4: Three Ward groups constrained by a symmetric eight-nearest-neighbor graph.

Table 4: Profiles of the three spatially constrained groups.

Group	Districts	Mean growth	Mean initial log light	Median density
1	252	0.1023	1.443	351.6
2	347	0.0406	1.999	495.5
3	41	0.0924	0.997	56.6

Group 1 combines relatively low initial light and fast growth; group 2 has higher initial light and slower growth; group 3 has very low density and fast growth. These profiles follow in part from the features used to form the groups.

6 Econometric theory and implementation

6.1 OLS benchmark and convergence

The absolute model regresses annual growth g_i on initial log light $z_{i,2013}$. The conditional model adds 2011 district characteristics and state indicators. HC3 robust standard errors allow heteroskedasticity. A negative coefficient is consistent with initially darker districts catching up in light, conditional on measured covariates. Under a restrictive transitional growth law,

$$\beta = \frac{\exp(-8\lambda) - 1}{8}, \quad \lambda = -\frac{\log(1 + 8\beta)}{8}$$

is an implied speed.

With state 01 as the reference, the estimated conditional equation is

$$\begin{aligned} \hat{g}_i = & 0.22351 - 0.04569 \text{ initial log light}_i \\ & - 0.01563 \text{ log density}_i + 0.01361 \text{ literacy}_i \\ & - 0.01720 \text{ agricultural worker share}_i - 0.01023 \text{ SC/ST share}_i + \widehat{\text{state effect}}_i. \end{aligned}$$

The intercept is a model reference term; the plotted line in Figure 2 is the simpler unconditional fit.

6.2 Three spatial alternatives

Let $\mathbf{Z}$ denote the design matrix and $\mathbf{W}$ the row-standardized spatial weights matrix. The three alternatives are

$$\text{SAR: } \mathbf{g} = \rho \mathbf{Wg} + \mathbf{Zb} + \varepsilon, \tag{4}$$

$$\text{SEM: } \mathbf{g} = \mathbf{Zb} + \mathbf{u}, \quad \mathbf{u} = \lambda \mathbf{Wu} + \varepsilon, \tag{5}$$

$$\text{SDM: } \mathbf{g} = \rho \mathbf{Wg} + \mathbf{Zb} + \mathbf{WZ}\theta + \varepsilon. \tag{6}$$

The simultaneous spatial-outcome term in SAR requires maximum likelihood; ρ captures conditional spatial association. SEM attributes dependence to unobserved spatial shocks. SDM additionally includes neighbors' initial light and other continuous characteristics. State indicators are never spatially lagged. All spatial models use Gaussian maximum likelihood with the Ord eigenvalue method.

In the SDM, impacts require

$$\mathbf{S}_k = (\mathbf{I} - \rho \mathbf{W})^{-1}(\beta_k \mathbf{I} + \theta_k \mathbf{W}).$$

The average direct impact is $\text{tr}(\mathbf{S}_k)/n$; the average total impact is $\mathbf{1}^\top \mathbf{S}_k \mathbf{1}/n$; and the indirect impact is total minus direct. Exact matrix calculations retain the zero-neighbor rows. Uncertainty intervals draw 5,000 times from the full model covariance matrix. They do not cover uncertainty about $\mathbf{W}$ or the source measurement.

6.3 Diagnostics

We compare AIC, a nested likelihood-ratio test for SAR versus SDM, and Moran tests on model residuals. OLS standard errors are HC3; spatial-model standard errors use the Gaussian likelihood. Neither specification delivers causal identification.

7 Model estimates and interpretation

Table 5: OLS and spatial-model estimates.

Model	Initial coefficient	Spatial parameter	AIC	Residual I
OLS: initial level only	−0.0574	—	—	—
OLS: controls + state	−0.0457	—	−2637.6	0.1693
SAR: controls + state	−0.0433	$\rho = 0.1966$	−2658.3	0.0690
SEM: controls + state	−0.0459	$\lambda = 0.3169$	−2672.9	0.0114*
SDM: controls + state	−0.0468	$\rho = 0.3253$	−2673.7	−0.0037

*SEM reports Moran’s I for its filtered innovations. All models use 640 districts.

The OLS conditional initial-light coefficient has HC3 SE 0.00344 and $R^2 = 0.800$. Its implied speed is 5.69% per year (approximately 4.67%–6.80% from its coefficient interval), conditional on the transitional-growth interpretation.

SAR leaves modest residual spatial dependence ($I = 0.0690$, $p = 0.024$). SEM filtered innovations ($I = 0.0114$, $p = 0.665$) and SDM residuals ($I = -0.0037$, $p = 0.935$) do not show detectable residual clustering. SDM has the lowest AIC, but its 0.84-point advantage over SEM is small. The nested SAR-versus-SDM likelihood-ratio statistic is 25.40 on five extra spatially lagged regressors ($p = 0.000117$). The data support additional spatial structure compared with SAR without decisively separating covariate spillovers from spatially correlated omitted factors.

Table 6: Spatial Durbin impacts of initial log light.

Impact	Point estimate	95% lower	95% upper
Average direct	−0.04652	−0.05089	−0.04221
Average indirect	0.00428	−0.00358	0.01222
Average total	−0.04225	−0.05107	−0.03359

The neighboring initial-light coefficient in the SDM is positive, but the average indirect-impact interval crosses zero. It is therefore not evidence for a precisely signed average spillover. The negative direct and total impacts align with convergence in this light proxy. The outcome-spatial parameter is an association, not proof that one district caused another to grow.

8 Robustness checks

Table 7: Alternative samples, periods, measures, and spatial weights.

Specification	Districts	Conditional OLS β (HC3 SE)	SAR ρ
VIIRS median-masked	640	−0.0458 (0.0035)	0.214
VIIRS 2014–2019	640	−0.0536 (0.0044)	0.252
VIIRS eight-neighbor graph	640	−0.0457 (0.0034)	0.141
Two-year-smoothed endpoints	640	−0.0445 (0.0036)	0.238
Queen non-islands	628	−0.0460 (0.0035)	0.184
Separate DMSP 2000–2013	640	−0.0259 (0.0026)	0.450
Trim 1% in both growth tails	626	−0.0421 (0.0032)	—

The negative initial-light association persists under alternative VIIRS masking, a pre-2020 interval, endpoint smoothing, geographic definitions, removal of Queen islands, and trimming. The DMSP result is a historical sign check only; its coefficients must not be compared numerically with VIIRS due to sensor and processing differences.

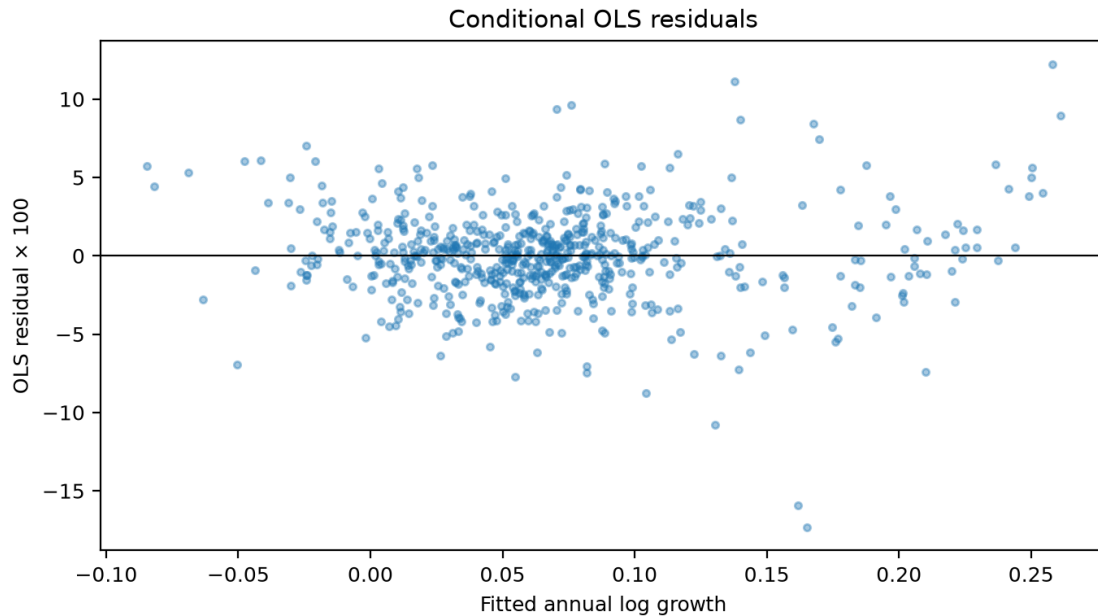


Figure 5: Conditional OLS residuals versus fitted annual log growth.

9 Limits and conclusion

First, this is an analysis of lights, not directly observed district GDP or consumption. Electrification, saturation, blooming, masking, and local economic composition can alter the lights–output relation. Second, fixed 2011 population does not measure migration or population growth by 2021. Third, state controls cannot remove district-specific omitted variables, common regional shocks, or measurement error. The growth formula reuses the initial light observation, so regression to the mean may contribute to its negative coefficient. Fourth, the spatial models rely on a chosen neighbor graph and Gaussian likelihood; geometry gaps produce disconnected Queen components.

The supported conclusions are narrower: normalized light rose in most districts; initially darker

districts tended to show faster subsequent light growth; and that growth was strongly spatially patterned. SEM and SDM explain more of the residual geographic pattern than a conventional regression. A stronger claim about economic convergence or causal spatial spillovers would require validated output or consumption measures, annual population, and a research design with exogenous sources of variation.

9.1 Reproducibility and data rights

The accompanying repository contains harmonized panels, district polygons, numerical tables, figures, the Python pipeline, source links, and LaTeX source. SHRUG-derived data and maps retain the upstream noncommercial CC BY-NC-SA 4.0 terms. Data lineage and exact variable definitions appear in `DATA_SOURCES.md` in the full analysis repository.

References

- [1] Asher, S., Lunt, T., Matsuura, R., and Novosad, P. (2021). Development research at high geographic resolution: An analysis of night lights, firms, and poverty in India using the SHRUG open data platform. *World Bank Economic Review*, 35(4), 845–871. doi:10.1093/wber/lhab003.
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